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# Safety data sheet



BASF safety data sheet. This is a translation of the country-specific safety data sheet into a language other than that required by law. This document does not replace the safety data sheet provided according to Regulation (EC) No 1907/2006.

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Product: **Rak 2 Ultra**

(ID no. 30682515/SDS_CPA_EU/EN)

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## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

# Rak 2 Ultra

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: crop protection product, pheromone

### 1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Crop Protection

Telephone: +49 621 60-27777
E-mail address: Produktinformation-Pflanzenschutz@basf.com

### 1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

## SECTION 2: Hazards Identification

### 2.1. Classification of the substance or mixture

For the classification of the mixture the following methods have been applied: extrapolation on the concentration levels of the hazardous substances, on basis of test results and after evaluation of experts. The methodologies used are mentioned at the respective test results.

According to Regulation (EC) No 1272/2008 [CLP]

| | |
|---|---|
| Skin Corr./Irrit. 2 | H315 Causes skin irritation. |

For the classifications not written out in full in this section the full text can be found in section 16.

### 2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

Pictogram:

Signal Word:

Warning

Hazard Statement:

| | |
|---|---|
| H315 | Causes skin irritation. |
| EUH401 | To avoid risks to human health and the environment, comply with the instructions for use. |

Precautionary Statement:

| | |
|---|---|
| P101 | If medical advice is needed, have product container or label at hand. |
| P102 | Keep out of reach of children. |
| P103 | Read carefully and follow all instructions. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P264 | Wash contaminated body parts thoroughly after handling. |
| P280 | Wear protective gloves/clothing/eye protection. |

Precautionary Statements (Response):

| | |
|---|---|
| P303 + P352 | IF ON SKIN (or hair): Wash with plenty of soap and water. |
| P332 + P313 | If skin irritation occurs: Get medical attention. |
| P362 + P364 | Take off contaminated clothing and wash it before reuse. |

Hazard determining component(s) for labelling: E7,E/Z9-Dodecadienylacetate

### 2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

See section 12 - Results of PBT and vPvB assessment.

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.
Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

## SECTION 3: Composition/Information on Ingredients

### 3.1. Substances

Not applicable

### 3.2. Mixtures

Chemical nature

crop protection agent in dispenser/ampoula, pheromone, Vapor releasing product (VP)
Chamber 1 contains:stabilized blend of 7,9-Dodecadien-1-ol, 1-acetate and Dodecylacetat: 400 mg

Regulatory relevant ingredients

E7,E/Z9-Dodecadienylacetate

| | |
|---|---|
| Content (W/W): 4.39 % | Skin Corr./Irrit. 2 |
| CAS Number: 54364-62-4 | Aquatic Chronic 3 |
| | H315, H412 |

Dodecyl acetate

| | |
|---|---|
| Content (W/W): 2.71 % | Aquatic Chronic 4 |
| CAS Number: 112-66-3 | H413 |
| EC-Number: 203-995-1 | |

2,6-di-tert-Butyl-p-cresol

| | |
|---|---|
| Content (W/W): < 1 % | Aquatic Acute 1 |
| CAS Number: 128-37-0 | Aquatic Chronic 1 |
| EC-Number: 204-881-4 | M-factor chronic: 1 |
| REACH registration number: 01-2119555270-46, 01-2119565113-46 | H400, H410 |

polyethylene

Content (W/W): < 95 %
CAS Number: 9002-88-4

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

## SECTION 4: First-Aid Measures

### 4.1. Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

### 4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

### 4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## SECTION 5: Fire-Fighting Measures

### 5.1. Extinguishing media

Suitable extinguishing media:
water spray, carbon dioxide, foam, dry powder

Unsuitable extinguishing media for safety reasons:
water jet

### 5.2. Special hazards arising from the substance or mixture

Endangering substances: carbon monoxide, Carbon dioxide, nitrogen oxides, metal oxides
Advice: The substances/groups of substances mentioned can be released in case of fire.

### 5.3. Advice for fire-fighters

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
In case of fire and/or explosion do not breathe fumes. Keep containers cool by spraying with water if exposed to fire. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

## SECTION 6: Accidental Release Measures

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Avoid contact with the skin, eyes and clothing.

### 6.2. Environmental precautions

Do not discharge into the subsoil/soil. Do not discharge into drains/surface waters/groundwater.

### 6.3. Methods and material for containment and cleaning up

For small amounts: Sweep/shovel up.
For large amounts: Sweep/shovel up.
Dispenser/ampoules should be collected. Contain and collect mechanically and dispose of in accordance with local regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations.

### 6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

## SECTION 7: Handling and Storage

### 7.1. Precautions for safe handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
Keep away from sources of ignition - No smoking. The product is combustible.

### 7.2. Conditions for safe storage, including any incompatibilities

Segregate from foods and animal feeds. Odour-releasing: Segregate from products sensitive to odours.
Further information on storage conditions: Keep in a cool place. Keep at temperature not exceeding 5 °C.

Protect from temperatures above:5 °C
Diffusion if stored for extended periods above the indicated temperature.

### 7.3. Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

## SECTION 8: Exposure Controls/Personal Protection

### 8.1. Control parameters

Components with occupational exposure limits

9002-88-4: polyethylene

### 8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Respiratory protection not required.

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

General safety and hygiene measures

The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

## SECTION 9: Physical and Chemical Properties

### 9.1. Information on basic physical and chemical properties

| | |
|---|---|
| State of matter: | liquid |
| Form: | liquid |
| Colour: | light yellow clear |
| Odour: | sweetish, faint odour |
| Odour threshold: | Not determined due to potential health hazard by inhalation. |

| | |
|---|---|
| Melting point: | approx. -25 °C<br>The data given are those of the active ingredient. |
| Boiling point: | approx. 298 °C<br>The data given are those of the active ingredient. |
| Flammability: | not applicable |
| Lower explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. |
| Upper explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. |
| Flash point: | 137 °C |
| Auto-ignition temperature: | approx. 254 °C (Directive 92/69/EEC, A.15) |
| Thermal decomposition: | 140 °C, 10 kJ/kg (DSC (OECD 113))<br>285 °C, > 210 kJ/kg (DSC (OECD 113))<br>Not a substance liable to self-decomposition according to UN transport regulations, class 4.1. |
| pH value: | approx. 6 - 8<br>(20 °C)<br>The data given are those of the active ingredient. |
| Viscosity, dynamic: | approx. 4.5 mPa.s |
| Solubility in water: | insoluble, Information based on the main component/s. |
| Partitioning coefficient n-octanol/water (log Kow): | The statements are based on the properties of the individual components. |

*Information on: Dodecyl acetate*
*Partitioning coefficient n-octanol/water (log Kow): 5.78*
---

| | |
|---|---|
| Vapour pressure: | approx. 0.017 Pa<br>(approx. 20 °C)<br>The data given are those of the active ingredient. |
| Relative density: | 0.888<br>(20 °C) |
| Density: | approx. 0.89 g/cm3<br>(approx. 20 °C) |

Relative vapour density (air):
not applicable

### 9.2. Other information

**Information with regard to physical hazard classes**

Explosives

Explosion hazard: not explosive

Oxidizing properties

Fire promoting properties: not fire-propagating

**Other safety characteristics**

Other Information: The product consists of a dispenser. Stated information applies for the dispenser content if not differently indicated.

Evaporation rate:
not applicable

## SECTION 10: Stability and Reactivity

### 10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

### 10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated.

### 10.3. Possibility of hazardous reactions

No hazardous reactions if stored and handled as prescribed/indicated.

### 10.4. Conditions to avoid

See SDS section 7 - Handling and storage.

### 10.5. Incompatible materials

Substances to avoid:
strong bases, strong acids, strong oxidizing agents

### 10.6. Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

## SECTION 11: Toxicological Information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Assessment of acute toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact. Virtually nontoxic by inhalation.

Experimental/calculated data:
LD50 rat (oral): > 5,000 mg/kg (OECD Guideline 401)
No mortality was observed.

*Information on: E7,E/Z9-Dodecadienylacetate*
*Experimental/calculated data:*
*LC50 (by inhalation): > 5.29 mg/l 4 h*
*No mortality was observed.*
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*Information on: E7,E/Z9-Dodecadienylacetate*
*Experimental/calculated data:*
*rat (dermal): > 2,000 mg/kg*
*The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
---

Irritation

Assessment of irritating effects:
Skin contact causes irritation. Not irritating to the eyes. The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: E7,E/Z9-Dodecadienylacetate*
*Experimental/calculated data:*
*Skin corrosion/irritation*
*rabbit: Irritant. (OECD Guideline 404)*
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*Information on: E7,E/Z9-Dodecadienylacetate*
*Experimental/calculated data:*
*Serious eye damage/irritation*
*rabbit: non-irritant (OECD Guideline 405)*
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Respiratory/Skin sensitization

Assessment of sensitization:
There is no evidence of a skin-sensitizing potential. The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: E7,E/Z9-Dodecadienylacetate*
*Experimental/calculated data:*
*Guinea pig maximization test :* Skin sensitizing effects were not observed in animal studies.
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Germ cell mutagenicity

Assessment of mutagenicity:
No data required for Straight Chained Lepidopteran Pheromones (SCLP) following the recommendation of OECD [OECD Series on Pesticides, No. 12 - Guidance for Registration Requirements for Pheromones and Other Semiochemicals Used for Arthropod Pest Control, 26-Feb-2002 (ENV/JM/MONO (2001)12)]

Carcinogenicity

Assessment of carcinogenicity:
No data required for Straight Chained Lepidopteran Pheromones (SCLP) following the recommendation of OECD [OECD Series on Pesticides, No. 12 - Guidance for Registration Requirements for Pheromones and Other Semiochemicals Used for Arthropod Pest Control, 26-Feb-2002 (ENV/JM/MONO (2001)12)]

Reproductive toxicity

Assessment of reproduction toxicity:
No data required for Straight Chained Lepidopteran Pheromones (SCLP) following the recommendation of OECD [OECD Series on Pesticides, No. 12 - Guidance for Registration Requirements for Pheromones and Other Semiochemicals Used for Arthropod Pest Control, 26-Feb-2002 (ENV/JM/MONO (2001)12)]

Developmental toxicity

Assessment of teratogenicity:
No data required for Straight Chained Lepidopteran Pheromones (SCLP) following the recommendation of OECD [OECD Series on Pesticides, No. 12 - Guidance for Registration Requirements for Pheromones and Other Semiochemicals Used for Arthropod Pest Control, 26-Feb-2002 (ENV/JM/MONO (2001)12)]

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
No data required for Straight Chained Lepidopteran Pheromones (SCLP) following the recommendation of OECD [OECD Series on Pesticides, No. 12 - Guidance for Registration Requirements for Pheromones and Other Semiochemicals Used for Arthropod Pest Control, 26-Feb-2002 (ENV/JM/MONO (2001)12)]

Aspiration hazard

No aspiration hazard expected.
The product has not been tested. The statement has been derived from the properties of the individual components.

Interactive effects

No data available.

## 11.2. Information on other hazards

Endocrine disrupting properties

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

Other information

Other relevant toxicity information

Misuse can be harmful to health.

## SECTION 12: Ecological Information

### 12.1. Toxicity

Assessment of aquatic toxicity:
At the present state of knowledge, no negative ecological effects are expected.

Toxicity to fish:
Study not necessary due to exposure considerations.

Aquatic invertebrates:
Study not necessary due to exposure considerations.

Aquatic plants:
Study not necessary due to exposure considerations.

### 12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
Easily eliminated from water. Can be eliminated mechanically.

### 12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on:Dodecyl acetate*
*Bioaccumulation potential:*
*Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is possible.*
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### 12.4. Mobility in soil

Assessment transport between environmental compartments:
Volatility: The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on:E7,E/Z9-Dodecadienylacetate*
*Assessment transport between environmental compartments:*
*Volatility: The substance will rapidly evaporate into the atmosphere from the water surface.*
*Adsorption in soil: Study not necessary due to exposure considerations.*

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### 12.5. Results of PBT and vPvB assessment

The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

### 12.6. Endocrine disrupting properties

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

### 12.7. Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

### 12.8. Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

## SECTION 13: Disposal Considerations

### 13.1. Waste treatment methods

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

## SECTION 14: Transport Information

**Land transport**

ADR

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |

| | |
|---|---|
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

RID

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user: | None known |

Transport in inland waterway vessel
Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Air transport**

IATA/ICAO

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

### 14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

### 14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

### 14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

### 14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

### 14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

### 14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

### 14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

## SECTION 15: Regulatory Information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Prohibitions, Restrictions and Authorizations

Annex XVII of Regulation (EC) No 1907/2006: Number on List: 3

Restrictions of Regulation (EC) No 1907/2006, Annex XVII, do not apply for the intended use(s) of the product given in this SDS.

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU): Listed in above regulation: no

To avoid risks to man and the environment, comply with the instructions for use.

### 15.2. Chemical Safety Assessment

Advice on product handling can be found in sections 7 and 8 of this safety data sheet.

## SECTION 16: Other Information

For proper and safe use of this product, please refer to the approval conditions laid down on the product label.

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

| | |
|---|---|
| Skin Corr./Irrit. | Skin corrosion/irritation |
| Aquatic Chronic | Hazardous to the aquatic environment - chronic |
| Aquatic Acute | Hazardous to the aquatic environment - acute |
| H315 | Causes skin irritation. |
| EUH401 | To avoid risks to human health and the environment, comply with the instructions for use. |
| H412 | Harmful to aquatic life with long lasting effects. |
| H413 | May cause long lasting harmful effects to aquatic life. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.